

Individual Reflection

1. If you did it as a group, how were tasks distributed, and what was your specific task? What challenges did you encounter?

In our group (Group-04), we divided the workload following the MLOps lifecycle. My main job was to oversee the technical validation of the entire pipeline to make sure everything actually worked together. A big part of my role was also evaluating the project's feasibility; I had to think critically about whether our code could handle real-world data.

2. What challenges did you encounter?

The most significant challenge was the project selection and time management. Our team initially evaluated four different project concepts before collectively deciding on the "AI-Powered Hyper-Personalized Shopping Assistant." This extensive deliberation, while ensuring a high-quality project direction, left us with a very tight schedule for the actual implementation. Additionally, as a student, navigating the integration of multiple cloud services was a steep learning curve; many technical configurations remained complex and unclear during the initial stages, requiring extra effort to debug and synchronize the various components of the pipeline.

3. Explain in your own words the advantages and disadvantages of using cloud services for implementing and deploying machine learning applications. You can use the application you selected for this project as a reference in your answer.

Advantages: The primary advantage of cloud services, such as Google Cloud Platform (GCP) used in our Shopping Assistant project, is scalability and efficiency. Tools like Vertex AI and AutoML abstract the underlying infrastructure, allowing us to train and deploy models without managing physical servers. This "serverless" environment significantly accelerates the development cycle. Furthermore, the integrated ecosystem (e.g., BigQuery for data, Cloud Run for deployment) ensures seamless data flow and provides robust built-in monitoring tools that would be difficult to build from scratch in an on-premise environment.

Disadvantages: On the other hand, the main disadvantages are technical complexity and cost management. The vast array of services and permission settings (IAM) can lead to a high barrier to entry for beginners, as reflected in the "unclear" challenges I faced during setup. Additionally, while the cloud offers flexibility, it can lead to vendor lock-in, making it difficult to migrate our Shopping Assistant to another provider without significant code refactoring. Lastly, without strict oversight, the operational costs of automated training and hosting can escalate quickly, especially when dealing with large datasets or high-frequency API calls.